

Tutorial–10: Solution Set

Course Details:

Course Code: 2025CSET106

Course Name: Discrete Mathematical Structures

Topics: Chinese Remainder Theorem, Permutations and Combinations, Inverse using Euclidean Algorithm

Part A: Chinese Remainder Theorem

Question 1 (a)

Solve the system of congruences:

$$x \equiv 5 \pmod{7}, \quad x \equiv 4 \pmod{9}$$

Solution:

Given $a_1 = 5$, $m_1 = 7$ and $a_2 = 4$, $m_2 = 9$.

Since $\gcd(7, 9) = 1$, Chinese Remainder Theorem is applicable.

$$M = m_1 m_2 = 7 \times 9 = 63$$

$$M_1 = M / m_1 = 9, \quad M_2 = M / m_2 = 7$$

Find inverses:

$$9 \equiv 2 \pmod{7}, \quad 2 \times 4 \equiv 1 \pmod{7} \Rightarrow M_1^{-1} = 4$$

$$7 \times 4 = 28 \equiv 1 \pmod{9} \Rightarrow M_2^{-1} = 4$$

$$x \equiv a_1 M_1 M_1^{-1} + a_2 M_2 M_2^{-1} \pmod{M}$$

$$x \equiv 5 \times 9 \times 4 + 4 \times 7 \times 4 = 292 \pmod{63}$$

$$x \equiv 40 \pmod{63}$$

Final Answer: $x \equiv 40 \pmod{63}$

Question 1 (b)

Solve the system:

$$x \equiv 1 \pmod{2}, \quad x \equiv 2 \pmod{3}, \quad x \equiv 3 \pmod{7}$$

Solution:

$$\text{Given } a_1 = 1, m_1 = 2; a_2 = 2, m_2 = 3; a_3 = 3, m_3 = 7.$$

$$M = 2 \times 3 \times 7 = 42$$

$$M_1 = 21, \quad M_2 = 14, \quad M_3 = 6$$

Find inverses:

$$21 \equiv 1 \pmod{2} \Rightarrow M_1^{-1} = 1$$

$$14 \equiv 2 \pmod{3}, \quad 2 \times 2 \equiv 1 \pmod{3} \Rightarrow M_2^{-1} = 2$$

$$6 \times 6 = 36 \equiv 1 \pmod{7} \Rightarrow M_3^{-1} = 6$$

$$x \equiv 1 \times 21 \times 1 + 2 \times 14 \times 2 + 3 \times 6 \times 6 \pmod{42}$$

$$x \equiv 185 \pmod{42}$$

$$x \equiv 17 \pmod{42}$$

$$\text{Final Answer: } x \equiv 17 \pmod{42}$$

Question 2

Solve the system:

$$x \equiv 2 \pmod{3}, \quad x \equiv 1 \pmod{4}, \quad x \equiv 1 \pmod{5}$$

Solution:

$$\text{Given } a_1 = 2, m_1 = 3; a_2 = 1, m_2 = 4; a_3 = 1, m_3 = 5.$$

$$M = 3 \times 4 \times 5 = 60$$

$$M_1 = 20, \quad M_2 = 15, \quad M_3 = 12$$

Find inverses:

$$20 \equiv 2 \pmod{3}, \quad 2 \times 2 \equiv 1 \pmod{3} \Rightarrow M_1^{-1} = 2$$

$$15 \equiv 3 \pmod{4}, \quad 3 \times 3 \equiv 1 \pmod{4} \Rightarrow M_2^{-1} = 3$$

$$12 \equiv 2 \pmod{5}, 2 \times 3 \equiv 1 \pmod{5} \Rightarrow M_3^{-1} = 3$$

$$x \equiv 2 \times 20 \times 2 + 1 \times 15 \times 3 + 1 \times 12 \times 3 \pmod{60}$$

$$x \equiv 161 \pmod{60}$$

$$x \equiv 41 \pmod{60}$$

Final Answer: $x \equiv 41 \pmod{60}$

Part B: Selection and Committee Formation

Question 3:

(a) Selecting 3 students from 10.

$$\text{Number of ways} = {}^{10}C_3 = 10! / (3!7!) = 120.$$

(b) Committee of 4 students from 10.

$$\text{Number of ways} = {}^{10}C_4 = 10! / (4!6!) = 210.$$

(c) President, Secretary, and Treasurer from 10 students.

Order matters, so permutations are used.

$$\text{Number of ways} = {}^{10}P_3 = 10 \times 9 \times 8 = 720.$$

(d) Selecting 2 boys and 2 girls from 6 boys and 4 girls.

$$\text{Number of ways} = {}^6C_2 \times {}^4C_2 = 15 \times 6 = 90.$$

(e) Committee of at least one student from 10.

$$\text{Total subsets} = 2^{10} = 1024.$$

$$\text{Excluding empty set, number of committees} = 1023.$$

Part C: Linear and Circular Arrangements

Question 4:

(a) Arranging 8 students in a row: $8! = 40320$.

(b) Arranging 5 out of 8 students: ${}^8P_5 = 6720$.

(c) Circular arrangement of 8 students: $(8 - 1)! = 5040$.

(d) If 3 students are always together: Treat them as one unit.

$$\text{Arrangements} = 6! \times 3! = 4320.$$

(e) Two students never together:

$$\text{Required} = \text{Total} - \text{Together} = 8! - (7! \times 2!) = 30240.$$

Part D: Alphabet-Based Permutations

Question 5:

Alphabets: A, B, C, D, E, F, G (Vowels: A, E).

(a) 3-letter words without repetition: $7P3 = 210$.

(b) 4-letter words with repetition allowed: $7^4 = 2401$.

(c) 3-letter words starting with a vowel: $2 \times 6P2 = 60$.

(d) 4-letter words with exactly two vowels: $2C2 \times 5C2 \times 4! = 240$.

(e) 5-letter words with A and E together: $5P4 \times 2! = 240$.

Part E: Inverse Using Euclidean Algorithm

Question 6

a) Find the multiplicative inverse of 7 modulo 26 using the Euclidean Algorithm.

Answer

We want x such that:

$$7x \equiv 1 \pmod{26}$$

Apply Euclidean Algorithm:

$$26 = 7 \times 3 + 5$$

$$7 = 5 \times 1 + 2$$

$$5 = 2 \times 2 + 1$$

$$2 = 1 \times 2 + 0$$

Now back-substitute:

$$1 = 5 - 2 \times 2$$

$$2 = 7 - 5 \times 1$$

$$1 = 5 - (7 - 5) \times 2 = 3 \times 5 - 2 \times 7$$

$$5 = 26 - 7 \times 3$$

$$1 = 3(26 - 7 \times 3) - 2 \times 7$$

$$1 = 3 \times 26 - 11 \times 7$$

Thus:

$$-11 \times 7 \equiv 1(\text{mod}26)$$

Inverse of 7 modulo 26 is: 15 (since $-11 \equiv 15(\text{mod}26)$)

b) Find the multiplicative inverse of 11 modulo 17 using the Extended Euclidean Algorithm.

Answer

We want:

$$11x \equiv 1(\text{mod}17)$$

Euclidean Algorithm:

$$17 = 11 \times 1 + 6$$

$$11 = 6 \times 1 + 5$$

$$6 = 5 \times 1 + 1$$

$$5 = 1 \times 5 + 0$$

Back-substitution:

$$1 = 6 - 5$$

$$5 = 11 - 6$$

$$1 = 6 - (11 - 6) = 2 \times 6 - 11$$

$$6 = 17 - 11$$

$$1 = 2(17 - 11) - 11 = 2 \times 17 - 3 \times 11$$

Thus:

$$-3 \times 11 \equiv 1 \pmod{17}$$

Inverse of 11 modulo 17 is: 14

c) Find the multiplicative inverse of 17 modulo 43 using the Euclidean Algorithm.

Answer

We want:

$$17x \equiv 1 \pmod{43}$$

Euclidean Algorithm:

$$43 = 17 \times 2 + 9$$

$$17 = 9 \times 1 + 8$$

$$9 = 8 \times 1 + 1$$

$$8 = 1 \times 8 + 0$$

Back-substitution:

$$1 = 9 - 8$$

$$8 = 17 - 9$$

$$1 = 9 - (17 - 9) = 2 \times 9 - 17$$

$$9 = 43 - 17 \times 2$$

$$1 = 2(43 - 17 \times 2) - 17$$

$$1 = 2 \times 43 - 5 \times 17$$

Thus:

$$-5 \times 17 \equiv 1 \pmod{43}$$

Inverse of 17 modulo 43 is: 38
